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Selective attention event-related potential effects from auditory novel stimuli in children and adults

Sara Määttä[a,*], Ari Pääkkönen[a], Pia Saavalainen[b,c], Juhani Partanen[a]

[a]Department of Clinical Neurophysiology, Kuopio University Hospital, University of Kuopio, P.O. Box 1777, 70211 Kuopio, Finland
[b]Department of Psychology, University of Joensuu, Joensuu, Finland
[c]Department of Otorhinolaryngology, Division of Phoniatrics, Kuopio University Hospital, University of Kuopio, Kuopio, Finland

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Abstract

Objective: We investigated differences between children and adults in selective attention.

Methods: Event-related potentials of 9 year-old children and adults were studied. Subjects performed an active dichotic novelty oddball task. We examined age-related differences in early selection by comparing non-target tones and late selection by comparing target tones in the attended and unattended channels.

Results: In children, an attention effect was seen on the N1 response to standard tones. For the targets, both children and adults displayed enhanced P3b amplitudes on the attended side, and in adults, an attention effect was also seen on the N2 response. In children, novelty-elicited N2 responses were larger to left ear stimuli irrespective of the direction of attention. Adults displayed enhanced novelty-elicited N2 amplitudes on the attended side.

Conclusions: Developmental changes occur both in early attentional selection and target detection. Children employed efficiently the mechanisms of early selection when processing standard stimuli, whereas their processes in relation to novel stimuli were attention-independent and even varied with ear. Adults were able to maintain their attentional focus in the presence of unexpected stimuli.

Significance: The results of this study contribute to elucidation of the development of selective attention.

Keywords: Event-related potentials; Selective attention; Children; Lateralization; Novelty

1. Introduction

The ability to focus on selected sensory inputs while ignoring irrelevant inputs is a critical feature of cognition. It requires an ability to differentiate between the relevant and irrelevant stimulus stream, select the relevant stream, inhibit processing of the irrelevant stream and sustain focused attention on the selected stream over some period of time. These aspects of cognition improve with age. The majority of these data have come from behavioural studies of same–different judgements, speeded classification, selective listening, incidental learning and dichotic listening (for a review, see Lane and Pearson, 1982). However, behavioural indices can at best provide only indirect evidence on age-related changes and have failed to address systematically the fundamental question of what is developing in the child that would account for the age-related differences.

Event-related potentials (ERPs) offer a psychophysiological method for studying human brain maturation associated with different aspects of perceptual and cognitive development, yielding information not available from behavioural studies. According to Kahneman and Treisman (1984), a distinction between experimental attention paradigms can be made on the basis of task requirements in terms of 'early selection' versus 'late selection'. In the case of early selection, attention serves to prevent or reduce the processing of irrelevant stimuli, whereas in the case of late selection, attention serves to select and facilitate

* Corresponding author. Tel.: +358-17-173-247; fax: +358-17-173-244.
E-mail address: sara.maatt@kuh.fi (S. Määttä).

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the response to relevant stimuli. The authors associated early selection with paradigms in which subjects are exposed to stimuli from relevant and irrelevant channels, and are asked to select the relevant channel for further processing. Late selection was associated with tasks in which subjects are asked to detect predesignated stimulus channels among non-target channels (Kahneman and Treisman, 1984). In ERP studies of attention, perhaps the most commonly used experimental approach is the active oddball paradigm. In this paradigm, typically two classes of stimuli are presented, one occurring frequently (standard) and the other occurring infrequently (target), and the subject is required to distinguish between the two stimuli and to respond to the stimuli that are pre-designated as targets. Thus, the active oddball paradigm enables examinination of psychophysiological expressions of late selection through the comparison of responses to target and non-target stimuli in the attended channel. Variations of this paradigm include the so-called novelty oddball paradigm, in which a third class of stimuli (novelty) are also presented intermixed with the standard and target stimuli, and a dichotic oddball paradigm, in which the stimuli are randomly interspersed between the two ears, the subject being instructed to direct his attention towards one ear and to ignore the stimuli arriving at the opposite ear. A dichotic oddball paradigm enables the examination of early selection by comparing responses to attended non-targets and non-attended non-targets.

Both early and late ERPs are modulated by attention. The general observation in dichotic selective attention paradigms is that the ERPs to standard tones in the attended ear are negatively shifted compared with the ERPs to the same tones in the same ear when ignored. This attention-related negativity is attributed to endogenous processing negativity (PN) and interpreted as an electrocortical reflection of early between-channel selection (Näätänen, 1992; Näätänen et al., 1978). The P3b elicited by target tones in the active oddball paradigm is taken as a psychophysiological manifestation of late selection. A major theoretical interpretation of the P3b component is that it reflects brain functions associated with context and memory updating, and the amount of both voluntary and involuntary attention allocated to the stimulus processing (Donchin and Coles, 1988; Polich, 1996). This context updating theory of the P3 is supported by studies that have shown that the P3 amplitude increases as the target probability in an oddball paradigm decreases (Duncan-Johnson and Donchin, 1977), and that larger P3 amplitudes are associated with superior memory performance (for a review, see Polich, 1996). Therefore, the P3 can be considered as a manifestation of central nervous system activity involved with the processing of new information when attention is engaged to update memory presentations (Polich, 1996). Although P3b is not exclusively a measure of selective attention, it is often used as a measure of allocation of attentional resources (Donchin and Coles, 1988). Also, target stimuli processing has been found to be associated with an enhanced N2. In this context, the N2 is taken to reflect the stimulus comparison that is involved in discrimination between target and non-target stimuli (Oades et al., 1997). It has also been suggested that the N2 and P3 components reflect the awareness of the subject that an unexpected event has occurred (Leppert et al., 2003).

Selective attention has been examined extensively in adult ERP studies but they are relatively few on selective attention in children (for a review, see Ridderinkhof and van der Stel, 2000). In addition, to our knowledge, developmental studies on children's attention have been concentrated mostly on voluntary and sustained attention. In general, children have shown clear signs of selective attention as indicated by enhanced N1/PN amplitudes to stimuli in the attended channel (Bartgis et al., 2003; Berman and Friedman, 1995; Loiselle and Stamm, 1980; Lovrich and Stamm, 1983; Oades et al, 1997), and also the P3 target mechanism is present at an early age (Brooker, 1980 cited in Donald, 1983; van der Stelt et al., 1998). Studies on the development of selective attention reveal age-related improvements on the development of early (between-channel) selection attentional processes (Brooker, 1980 cited in Donald, 1983; van der Stelt et al., 1998), and also target detection processes improve in efficiency (indicated by Pb3 latency; Courchesne, 1978; Goodin and Squires, 1978; Martin et al., 1988), and may be less demanding (indicated by P3b amplitude) as children grow older. In general, these results support the findings of previous behavioural studies. However, it is not known at which stage of attentional selection these changes occur. Also, even though children are more easily distracted by unexpected stimuli than adults, it is not known how such stimuli are processed in the attended and unattended channels.

The aim of this study was to examine age-related changes in selective attention using an active dichotic novelty oddball paradigm, which enabled the studying of both early (between-channel selection) and late (target detection) phases of attentional selection with between channel-selection (reflected by the early N1/PN) preceding target detection (reflected by the later P3b). We hypothetized that developmental changes occur both in early and in late attentional selection. Previous developmental investigations in this area have focused on responses to standard or target stimuli and provided little detail of the age-related changes to novel stimuli. Our second hypothesis was that adults are better able to maintain their focus of attention on the selected channel irrespective of the stimulus type.

2. Subjects and methods

2.1. Subjects

Twenty normal children (aged 8–9 years, mean 9.3 years, 4 boys) and 10 normal healthy adults (aged 22–28 years, mean 24.9 years, two male) were studied. The children were

Table 1
Standard scores in neuropsychological tests

Subtest	Mean (SD)
WISC-R	
Arithmetic	11.7 (1.26)
Digit span	8.7 (2.5)
Coding	12.4 (2.72)
NEPSY	
Statue	10.4 (3.3)
Tower	12.1 (2.6)
Knock and tap	9.1 (3.6)
Visual attention	11.8 (5.0)
Auditory attention and response set	15.3 (2.1)
Design fluency	11.7 (2.9)

Notes: The raw scores of the tests were converted into the standard scores according to the manual (Finnish standardization). Scores 8–12 represent the average and scores <7 represent the performance below average, SD in brackets.

volunteers recruited from a local elementary school and the adults were medical students from the University of Kuopio. None of the subjects reported a history of significant psychiatric or neurological disease or severe head injury, and none was taking any medication that affects the central nervous system. All subjects were right-handed. In children, the handedness was determined by the report of their teacher (teacher's questionnaire) and in adult subjects, by their own report. The children attended normal education, and according to their teacher, none of them had any learning difficulties. In order to obtain valid measures of attention and executive functions of the children, 3 subtests of the Wechsler Intelligence Test for Children-Revised (Wechsler, 1984) and 6 subtests of NEPSY (Korkman et al., 1997), were administered. All children performed on all attentional tests within the normal range. The results of the tests are presented in Table 1. Informed written consent to participate in the study was obtained from the subjects and the childrens' parents. The study was approved by the Kuopio University Hospital Research Ethical Committee.

2.2. Stimuli

The ERPs were elicited using a dichotic 3-tone oddball paradigm with 86.8% standard (800 Hz), 10.8% deviant/target (560 Hz) and 2.4% novel tones. The duration of standard and deviant tones was 84 ms, including 7 ms rise and fall times. The novelty tones were 6 different complex 100 ms tone bursts including 2 ms rise and fall times (Fig. 1). The order of the tones in the sequence and also their occurrence to right and left ears were semi-randomized. The number of stimuli was 520 and the inter-stimulus interval (ISI) was 1 s. The stimuli were delivered by headphones to the right and left ears separately at 50 dB above the hearing level. The hearing level of each subject was tested at the beginning of the recording session from both ears.

2.3. Recording conditions

The ERPs were recorded with a 64 channel electrode cap. All electrodes were referred to the right mastoid, and when analysed re-referred to the average of the potentials at electrode sites P9 and P10 in order to avoid the bias caused by the unilateral reference. Off-line calculations allow the referencing to any site or sites desired (Dien, 1998; Picton et al., 1980). We chose the average of the P9 and P10 because in the electrode cap used in this study they are the electrodes nearest to the mastoid processes which

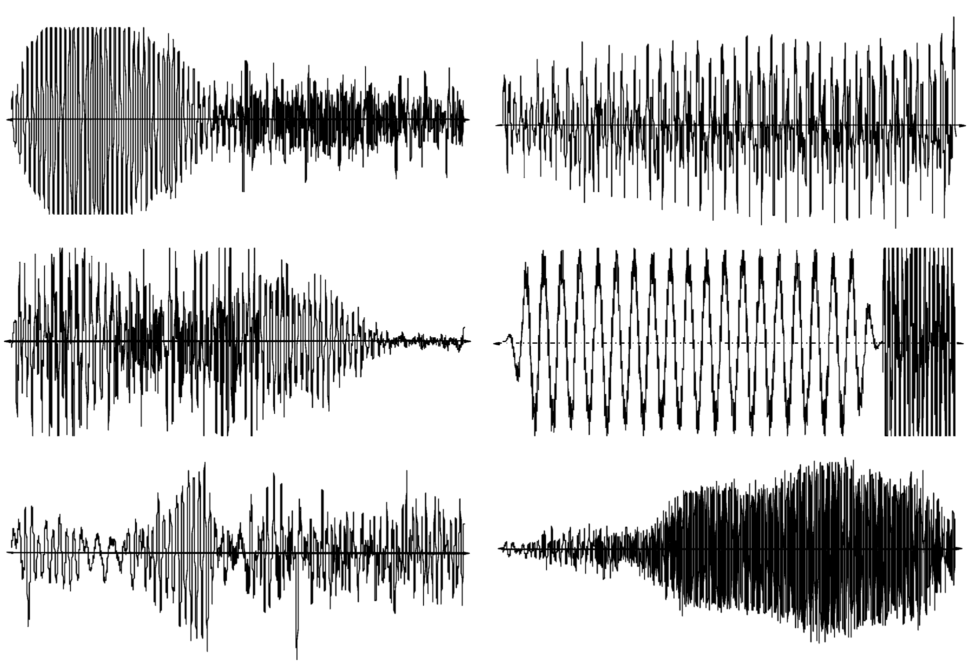

Fig. 1. The waveforms of novel stimuli. The novel tones were 6 different complex 100 ms tone bursts.

commonly serve as reference sites. Also, they were the furthest (most lateral) electrodes from the midline electrodes used in analysis, and they were the least active ones with respect to the processes dealt with in our study. Potentials reflecting vertical eye movements and eye blinks were recorded between electrodes placed above and below the right eye. Electrode-skin impedances were below 5 kΩ. Signals were amplified and digitised continuously at 250 Hz with a Neuroscan system (Neuroscan Inc., Virginia, USA). The data were transformed off-line to epochs of −100 to 900 ms relative to the onset of each stimulus. Epochs containing eye-movement artifacts were rejected (rejection levels +75 and −75 μV). The responses to standard, deviant and novelty stimuli were averaged separately for each subject. The averaged data were filtered digitally with high-pass cut-off at 0.1 Hz and low-pass cut-off frequency at 20 Hz (3 dB point of 24 dB/octave roll-off). The stimulus sequence was presented in 3 different recording conditions. In the first run, the subject was instructed to ignore the tones and to concentrate on watching a video (ignore condition). In the second and third run (the attended conditions), the subject was instructed to focus on the right or the left ear and to respond to each target stimuli on the attended side by pressing a button with their dominant (right) hand. Every other subject attended first to the left ear. In this report, the focus is on the attended conditions.

2.4. Data analysis

Peak amplitudes and latencies of the standard, target and novelty-elicited N1, N2 and P3 responses were measured with respect to the 100 ms prestimulus baseline in each subject from 5 midline electrodes (Fz, Fcz, Cz, Cpz, Pz) except for the P3 to targets, which in children was well-defined only at Pz, and therefore Pz was used for P3 measurements. An automatic peak detection method combined with manual checking was used. The N1 was defined as the maximum negative peak within 80–180 ms in children and 75–160 in adults, the N2 as the maximum negative peak within 200–290 ms in children and within 160–240 ms in adults. The novelty P3 was defined as the maximum positive peak in the latency range of 250–400 ms in children and 200–400 ms in adults, and target P3 as the maximum positivity between 250 and 500 ms in adults and 500 and 800 ms in children. These latency intervals were chosen on the basis of the location of peaks in children's and adults' grand-average waveforms.

Latency and amplitude data were analysed with repeated-measure analysis of variance (ANOVA) using SPSS for Windows 10.0 statistical program. A two-factor (two attended sides: right ear attended vs. left ear attended × two sides of stimulation: right ear vs. left ear) ANOVA was applied to the target P3 component, and a 3-factor (two attended sides: right ear attended vs. left ear attended × two sides of stimulation: right ear vs. left ear × 5 electrodes: Fz, Fcz, Cz, Cpz, Pz) ANOVA was applied to other ERP components separately in both age groups. The Greenhouse-Geisser correction was used (corrected P values are reported) where appropriate. Post-hoc paired comparisons were made using Tukey's HSD test with P set at 0.05.

When subjects detected a target tone in the attended ear they were required to respond by pressing a button. Pressing the button within 1000 ms interval after the target presentation was regarded as a hit, and the average reaction time (RT, in milliseconds) was computed for these trials. An incorrect button press during this interval was classified as an error and a trial with no response as a miss. Performance data analyses were carried out by means of ANOVAs separately for the RT data, number of hits and number of errors with group as the between-subject factor.

3. Results

3.1. Performance data

Table 2 shows the RT, the percentage of the correct responses and the number of errors in children and adults. The number of the hits was significantly smaller in children than in adults ($F(1,28)=27.932$, $P<0.001$). The children made also more errors in comparison with the adults ($F(1,28)=6.868$, $P=0.014$). Since the novelty-elicited ERP data in children demonstrated an asymmetry between the ears of presentation, we wanted to test whether this asymmetry was related to differences in the behavioural reactions to the novel stimuli between the ears. The only behavioural responses to novel stimuli that were recorded during the experiment were the false button presses, and a paired samples t test found no differences in the number of false button presses to the novel stimuli between the ears ($P=0.186$). Although the mean RT was longer in children than in adults, the difference was not significant ($P=0.093$). The attended side failed to reach statistical significance for

Table 2
Mean and SD of the RT (in ms) to hits, hit percentage, and number of errors in the performance of the target discrimination task for the children and adults groups

	Children	Adults
RT		
Targets to the left ear	580 (49.1)	542 (59.9)
Targets to the right ear	591 (66.2)	553 (67.7)
The percentage of correctly identified targets		
To the left ear	62.9% (21.3)	92.9% (8.1)
To the right ear	63.4% (17.7)	95.3% (5.6)
Number of errors		
Left ear	3.2 (2.8)	0.7 (1.2)
Right ear	5.0 (6.2)	1.0 (1.9)

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the RT, the number of hits or the number of errors in either group.

3.2. Amplitudes

3.2.1. Responses to standard tones

Fig. 2 shows ERP waveforms elicited by standard tones in response to left ear stimulation in children and adults at Fz, Cz and Pz. N1 was clearly observed in both groups, whereas only children displayed a prominent N2.

N1. In children, the repeated measures ANOVA yielded a significant side of stimulation (ear)×attended side interaction for the N1 amplitude ($F(1,19)=8.871$, $P=0.008$). Post-hoc comparisons with Tukey's HSD test revealed that the N1 amplitude was larger when the ear was attended than when it was not attended ($P<0.05$ for the right and the left ear) and in the attention left condition, tones to the left ear elicited larger N1 than tones to the right ear ($P<0.05$). The ANOVA also showed a significant main effect for the electrode site for both groups for the N1 amplitude ($F(4,76)=6.679$, $P=0.011$ in children; $F(4,36)=26.478$,

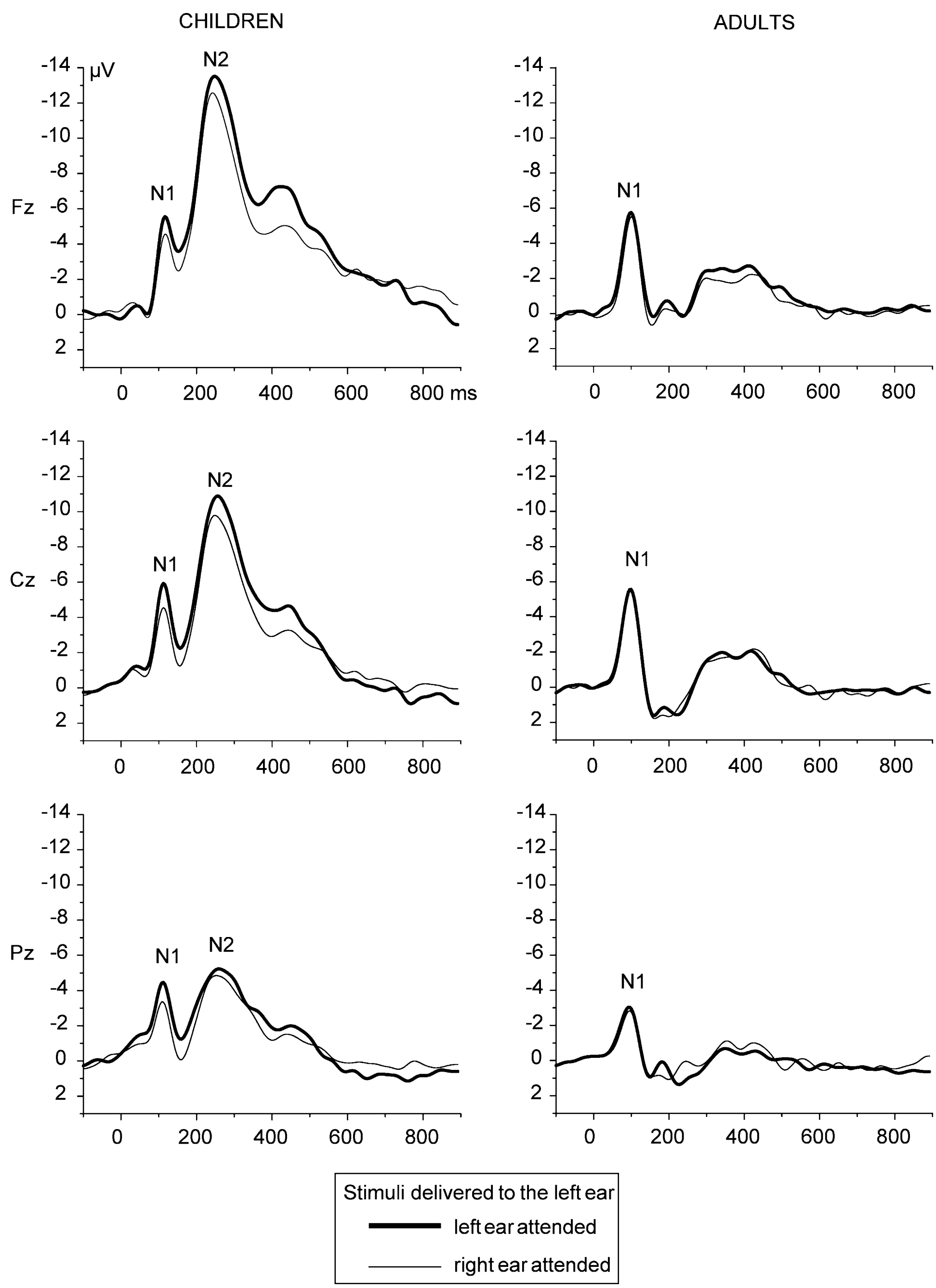

Fig. 2. Grand-average ERP waveforms at Fz, Cz and Pz elicited by standard stimuli presented to the left ear in children and in adults. In children, an attention effect was seen on the N1 response to standard tones.

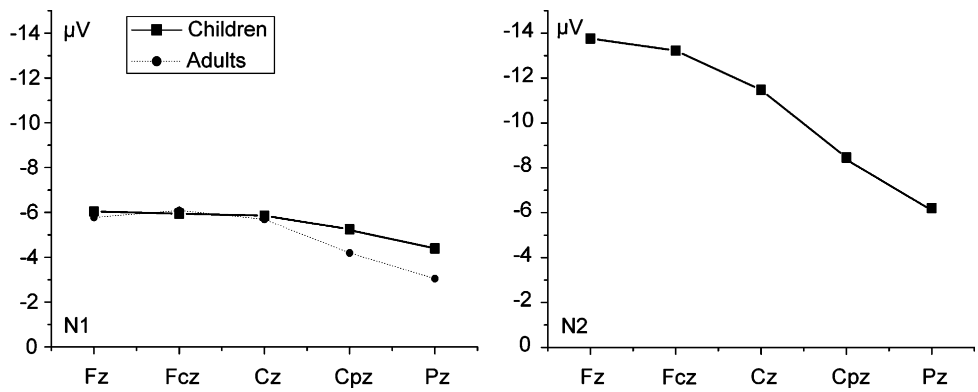

Fig. 3. N1 amplitudes in children and adults and N2 amplitudes in children in response to standard stimuli at 5 midline electrodes.

$P<0.001$ in adults). In children, the N1 amplitude was significantly smaller at Pz than at other electrode sites ($P<0.05$ in all cases). In adults, the N1 was significantly larger in amplitude at Fz, Fcz and Cz than at Cpz or Pz ($P<0.05$ in all cases; Fig. 3).

N2. In children, the ANOVA for the N2 amplitude yielded a significant ear×attended side interaction ($F(1,19)=7.433$, $P=0.013$). The results of Tukey's HSD tests indicated that tones presented to the left ear elicited greater N2 when the left ear was attended than when the right ear was attended ($P<0.05$), and that the N2 amplitude was larger on the attended side in comparison to the non-attended side ($P<0.05$ for the right and the left ear attended condition). A main effect for the electrode sites was also obtained ($F(4,76)=68.407$, $P<0.001$), post-hoc comparisons indicating that the N2 amplitude was significantly larger at Fz and Fcz that at other electrode sites ($P<0.05$ in all cases), at Cz significantly larger that at Cpz or Pz ($P<0.05$ in both cases), and and at Cpz significantly larger than at Pz ($P<0.05$; Fig. 3).

3.2.2. Responses to target tones

Grand-average ERP waveforms at Fz, Cz and Pz are shown in Fig. 4. In both age groups, the targets elicited a sequence of deflections comprised of N1, N2 and P3.

N1. The repeated measures ANOVA yielded a significant main effect for the electrode sites for both groups for the N1 amplitude ($F(4,76)=9.160$, $P=0.003$ in children; $F(4,36)=15.935$, $P=0.002$ in adults). The children displayed a frontal and the adults a frontocentral maximum. In children, the N1 amplitude was significantly smaller at Pz than at other electrode sites ($P<0.05$ in all cases). In adults, the N1 amplitude was significantly larger at Fz than at Pz, at Fcz larger than at Cpz and Pz, and at Cz larger than at Pz ($P<0.05$ in all cases; Fig. 5).

N2. In children, the direction of attention did not have significant effect on the N2 amplitudes whereas in adults, a significant ear×attended side interaction ($F(1,9)=15.405$, $P=0.003$) was obtained. Tukey's HSD test revealed that in adults, the N2 amplitude was larger when the ear was attended than when it was not attended ($P<0.05$ for the tones presented to the right and $P<0.05$ to the left ear). Also, in the attention right condition, tones to the right ear elicited larger responses than tones to the left ear ($P<0.05$), and in the attention left condition, tones to the left ear elicited larger responses than tones to the right ear ($P<0.05$). In both groups, the N2 amplitude data also showed a significant main effect of electrode site ($F(4,76)=45.696$, $P<0.001$ in children; $F(4.36)=20.224$, $P<0.001$ in adults). (Fig. 4b). In children, the N2 amplitude was significantly larger at Fz and Fcz that at other electrode sites ($P<0.05$ in all cases), at Cz significantly larger that at Cpz or Pz ($P<0.05$ in both cases), and and at Cpz significantly larger than at Pz ($P<0.05$). In adults, the N2 amplitude was significantly larger at Fz and Fcz that at other electrode sites ($P<0.05$ in all cases; Fig. 5).

P3. When subjects paid attention to stimuli, larger P3 amplitudes were recorded as compared to non-attended stimuli (ear×attended side ANOVA: ($F(1,19)=41.170$, $P<0.001$ in children; $F(1,9)=11.233$, $P=0.009$ in adults). Tukey's HSD test showed that P3 responses for the right ear were significantly larger when attended ($P<0.05$ in children and $P<0.05$ in adults) and P3 responses for the left ear were significantly larger when attended ($P>0.05$ in children and $P<0.05$ in adults). Also, in the attention right condition, tones to the right ear elicited larger responses than tones to the left ear ($P<0.05$ in both groups), and in the attention left condition, tones to the left ear elicited larger responses than tones to the right ear ($P<0.05$ in both groups).

3.2.3. Responses to novel tones

Fig. 6 shows the grand-average waveforms for the right ear attended and left ear attended novelty tones in children and adults in the midline electrodes. In both age groups, the novel tones elicited a sequence of deflections comprised of N1, N2 and P3.

N1. In children, no significant main effects or interactions were obtained. In adults, N1 amplitude data yielded a significant main effect for the electrode site ($F(4,36)=7.475$, $P=0.012$), the N1 amplitude being significantly smaller at Pz than at Fz, Fcz or Cz ($P<0.05$) (Fig. 7).

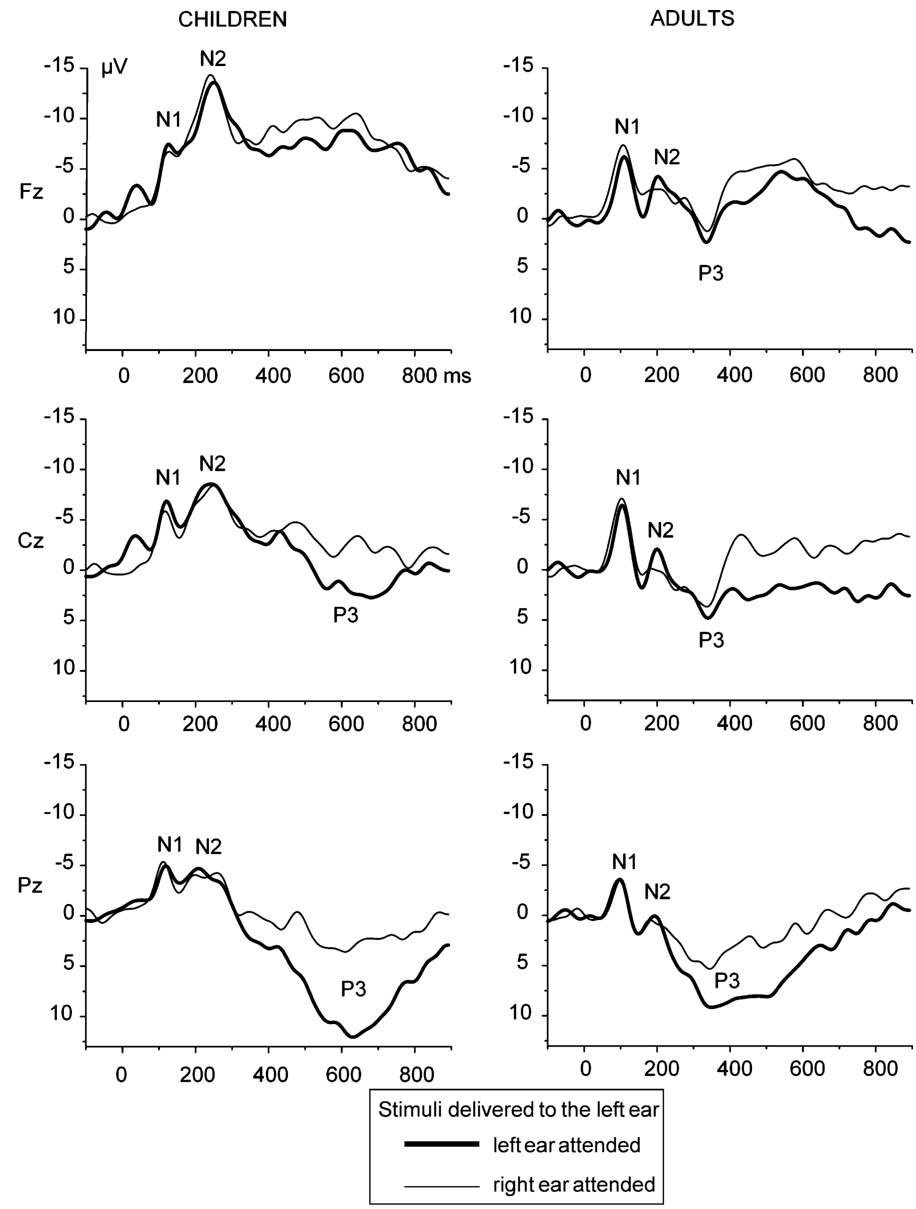

Fig. 4. Grand-average ERP waveforms at Fz, Cz and Pz elicited by target stimuli presented to the left ear in children and in adults. In adults, target tones presented to the attended ear elicited a response with a significantly larger amplitude than tones presented to the non-attended ear.

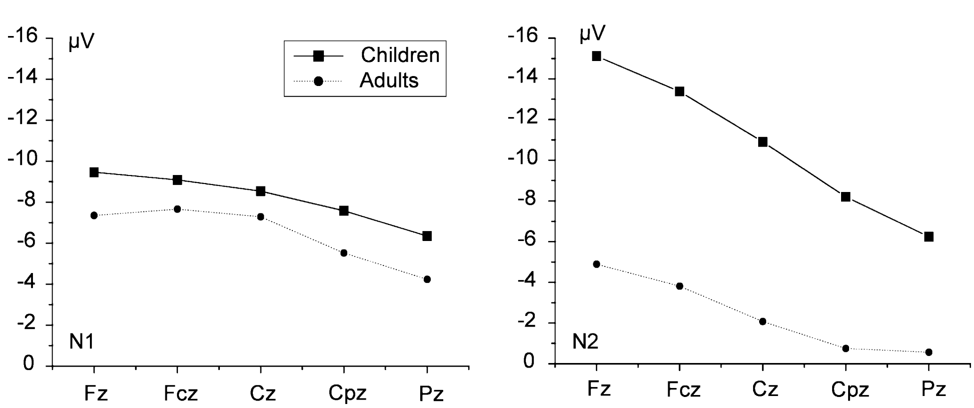

Fig. 5. N1 and N2 amplitudes in response to target stimuli in children and adults at 5 midline electrodes.

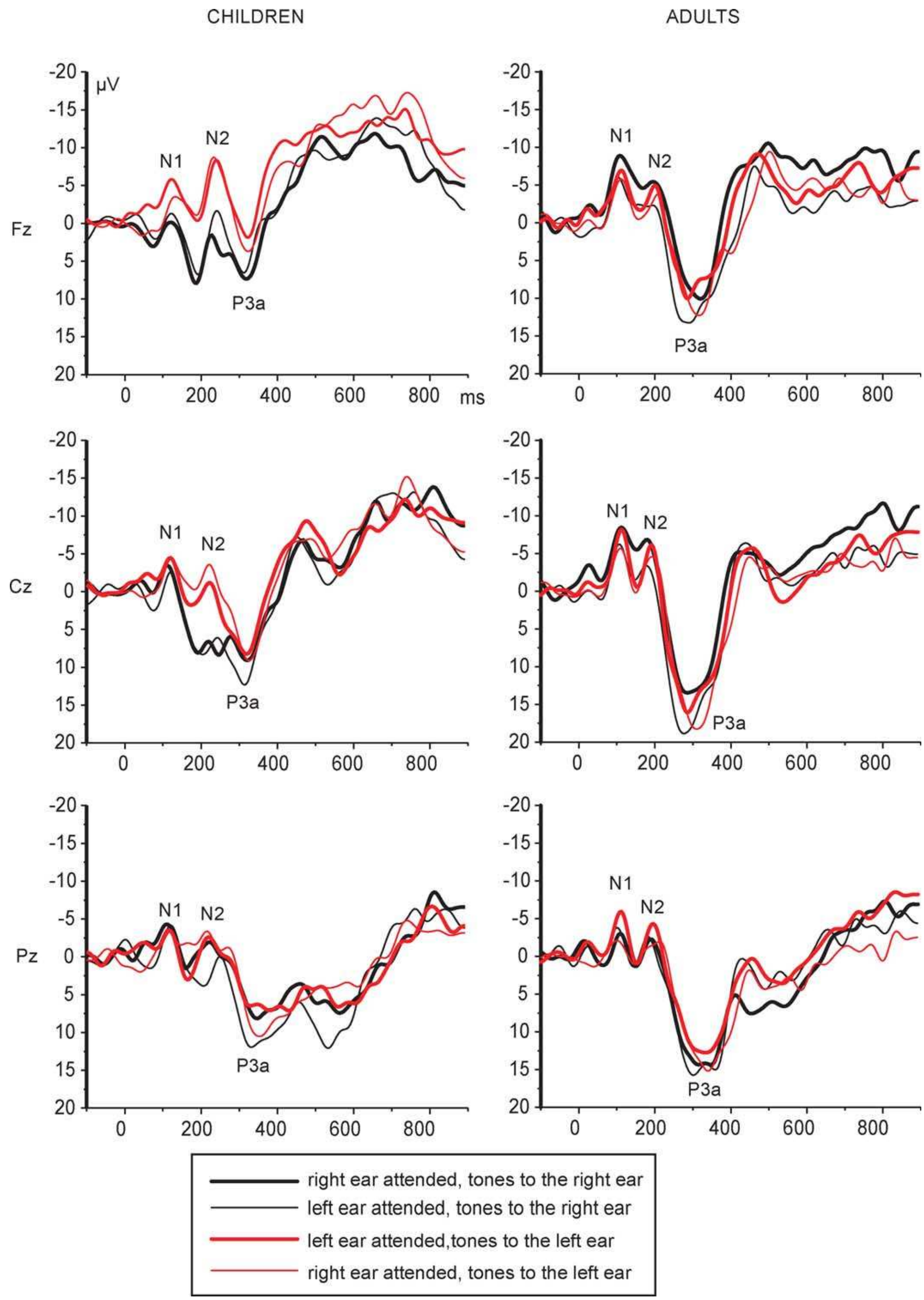

Fig. 6. Grand-average ERP waveforms at Fz, Cz and Pz elicited by novel stimuli in children and in adults. In adults, responses to novel tones delivered to the attended ear were significantly larger than those delivered to the non-attended ear, whereas in children, novelty-elicited N2 responses were larger to left ear stimuli irrespective of the direction of attention.

N2. In children, the amplitudes of the N2 responses were larger for the left ear stimuli irrespective of the direction of attention. The repeated measures ANOVA yielded a significant main effect for ear ($F(1,19)=5.286$, $P=0.033$), Tukey's HSD test indicating that tones presented to the left ear elicited a larger negativity in the N2 latency range than novelty tones presented to the right ear ($P<0.05$). The adult N2 amplitude data showed a significant ear $\times$ attended side interaction ($F(1,9)=6.663$, $P=0.03$), and Tukey's HSD test showed that adults were characterized by significantly larger N2 responses to the left ear tones when left ear was attended ($P<0.05$) and to the right ear tones to the when

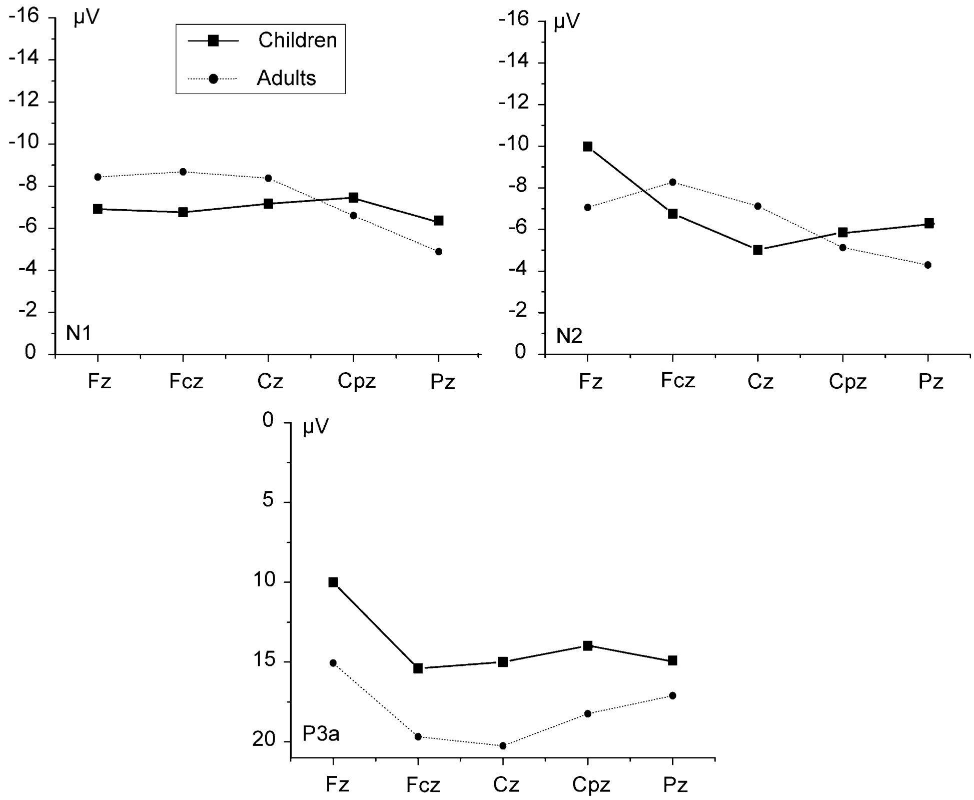

Fig. 7. N1, N2, and P3a amplitudes in response to novel stimuli in children and adults at 5 midline electrodes.

right ear was attended ($P<0.05$). In adults, the repeated measures ANOVA also yielded a significant main effect for the electrode site ($F(4,36)=7.860$, $P=0.006$) with N2 being significantly larger at Fzc than at Pz ($P<0.05$; Fig. 7).

P3a. In children and adults, the P3a amplitude data yielded a significant main effect for electrode ($F(4,76)=3.896$, $P=0.03$ in children; ($F(4,36)=4.786$, $P=0.033$ in adults). In children, the P3a was at its largest at a frontocentral electrode (Fcz>Fz, Cz, Cpz, and Pz, $P<0.05$ in all cases). In adults, Fz was significantly smaller at Fz than at Fcz, Cz or Cpz ($P<0.05$; Fig. 7).

Note. Since the number of novel stimuli was small, we applied a group×condition ANOVA on the number of novel stimuli after artifact rejection in each condition to make sure that signal-to-noise ratios were comparable across conditions and groups. No significant interactions (group×condition: $F<1$) or main effects (condition: $F<1$; group: $P=0.088$), were found indicating that the number of trials included in the analysis of ERPs was similar across the groups and the conditions.

3.3. Latency data

3.3.1. Standards

In children, the standard N1 latency data yielded a significant main effect for electrode ($F(4,76)=8.626$, $P<0.001$), N1 being significantly faster at Pz than at Fz, Fcz, and Cz, and significantly faster at Cpz than at Fz or Fcz ($P<0.05$ in all cases). Childrens latency data also yielded a significant ear×attended side interaction ($F(1,19)=5.834$, $P=0.026$). Tukey's HSD test revealed that the N1 latency was longer when the right ear was attended than when it was not attended ($P<0.05$). Also, in the attention right condition, tones to the right ear elicited slower responses than tones to the left ear ($P<0.05$).

3.3.2. Targets

In children, the N1 latency data yielded a significant main effect for electrode ($F(4,76)=14.902$, $P=0.001$), showing faster N1 at Pz compared with Fz or Fzc ($P<0.05$ in both cases). In adults, the N1 latency data yielded a significant ear×electrode interaction ($F(4,36)=4.782$, $P=0.017$), resulting from an earlier N1 response elicited by left ear stimulation than by right ear stimulation at Pz ($P<0.05$).

3.3.3. Novels

In children, the N1 latency data yielded a significant main effect for both electrode ($F(4,76)=4.197$, $P=0.018$) and ear ($F(1,19)=5.101$, $P=0.036$) showing earlier N1 responses for right ear stimulation ($P<0.05$) and significantly faster N1 at Pz compared with Fz ($P<0.05$). In children, the N2 latency data ANOVA also yielded a significant ear×attended side×electrode interaction ($F(4,76)=3.652$, $P=0.013$), but post-hoc testing failed to demonstrate significant main effects related to lateralization at any electrode site. In children and adults, the P3a latency data yielded a significant main effect for electrode ($F(4,76)=17.327$, $P<0.001$ in children and $F(4,36)=7.603$, $P<0.001$ in adults). In children, the P3a was significantly slower at Cpz and Pz compared with other electrode sites ($P<0.05$), and in adults, the P3a was significantly slower at Pz than at Fzc, Cz, or Cpz ($P<0.05$ in all cases).

4. Discussion

The results of this study add to the evidence that there are age-related changes in the ability to selectively attend to sensory input, and these changes can be seen in the responses to novel tones as well as to standard and target tones. We employed an active dichotic novelty oddball paradigm, which offered the opportunity to examine early between-channel selection in children and adults by comparing responses to non-target (standard and novelty) tones and late selection (target detection) by comparing responses to target tones in the attended and unattended channels. We observed N1 and N2 waves in response to standard tones in children and N1 in adults. Target-tones elicited N1, N2 and P3 and novel tones N1, N2 and P3a in both groups. The basic topographic features and the morphology of the ERPs found in this study are similar to those obtained at midline electrodes in previous studies and are not discussed further (Bruneau et al., 1997; Cycowicz and Friedman, 1997; Cycowicz et al., 1996; Takeshita et al., 2002).

The performance data suggested that the children found the paradigm more difficult than the adults as indicated by the smaller number of hits in children than in adults. The children made also more errors in than the adults—a finding that may be considered as an evidence of children using less strict criterion in discriminating targets from other tones. Although the mean RT was longer in children than in adults, the difference was not significant. The behavioural data failed to indicate any ear advantage in either group.

In children, standard tones presented to either ear elicited a greater N1 when the ear was attended than when it was not attended, and the attention effect was also seen in attention left -condition, where tones to the attended ear (left) elicited larger N1 than tones to the non-attended (right) ear. The N1 component of the auditory ERP, which is a relatively large negative deflection peaking at about 100 ms from stimulus onset, has played an important role in ERP research on selective attention (for a review, see Näätänen, 1992). The N1 is a transient response to sound onsets and offsets. It has multiple generators, most of them residing in the temporal lobe (for a review, see Näätänen and Picton, 1987; Woods, 1995). In children, the N1 latency is significantly longer than in adults, but the high susceptibility to the stimulus rate, the long refractory period and the frontocentral distribution implies a strong parallelism between the children's N1 and the adults' N1 (Ceponiene et al., 1998). In experimental conditions demanding selective attention, the attended standard stimuli are known to elicit enhanced N1 when compared to the N1 to the same stimuli when ignored (Hillyard et al., 1973), and this enhancement was attributed to an endogenous PN and interpreted as an electrocortical reflection of early between-channel selection (Näätänen, 1992; Näätänen et al., 1978). Also children have shown the effcts of selective attention on ERPs reflected by enhanced N1/PN amplitudes to stimuli in the attended channel (Bartgis et al., 2003; Berman and Friedman, 1995; Loiselle and Stamm, 1980; Lovrich and Stamm, 1983; Oades et al, 1997), but the discrimination processs necessary for separating the relevant and irrelevant channels seem to be more difficult and have the later onset the younger the child is (Berman and Friedman, 1995). In our study, the children but not the adults demonstrated attention effects on standard N1. This is consistent with the previous studies, in which a short ISI is generally considered to be neccessary for the N1 enhancement. With a slow stimulus rate (of the order of the 1 s) there is no need to be selectively attentive at the moment of stimulus presentation as there is adequate time to handle information in both stimulus channels afterwards (Hartley, 1970; Näätänen, 1990). With the 1s ISI used in this study, the adults could use the late-selection strategy. Thus, a condition, which in adults turned out to be a low-load one, resulted in the activation of selective attention mechanisms in children. In addition to N1, the attention effect was seen on childrens' N2 (N250) as well, which is childrens additional parallel way of processing non-target stimuli. N2 may represent a discriminative process (Johnstone et al., 1996), which is supposed to be automatic and preattentive (Takeshita et al., 2002).

For the targets, adults showed the effect of selective attention on both N2 and P3b, whereas in children, the N2 components were of equal amplitude for both attended and unattended targets, and attention effect was seen only on the P3b component. The target-elicited N2 component, a negative deflection peaking at about 200 ms, has been characterized as representing various subcomponents with different functions (Näätänen and Picton, 1986), which have been linked to early target recognition processes (Deacon et al., 1991; Ritter et al., 1983) and stimulus classification (Ritter et al., 1979, 1983) and stimulus deviance detection (Simson et al., 1977). The P3b wave to target stimuli occurring at about 300 ms is related to stimulus evaluation and is independent of response selection and execution processes (Picton, 1992). The enhancement of the P3b wave in response to attended stimuli is in accordance with previous studies. It is known that when subjects pay attention to stimuli in given modality, larger P3b waves are recorded as compared to non-attended stimuli (Hillyard et al., 1973), and this P3b mechanism is present also in children (Brooker, 1980 in Donald, 1983). ERP studies of attentional effects on target-elicited N2 are scarce. The data obtained here indicate that in adults, substantially large amount of available attentional resources were allocated to the attended tones in even before the P3b generating process. This was indexed as enhanced N2 waves in response to attended stimuli. In children, the N2 components of equal amplitude for both attended and unattended targets may suggest that the attended and unattended target stimuli are processed equally before P3b onset. The target N2 may thus represent a discriminative process which becomes more efficient with age.

Adults appear to be better able to selectively attend to the instructed ear and to maintain their attentional focus event in the presence of new, unexpected stimuli, whereas children show attention-independent automatic processing of such stimuli. This was indexed as enhanced N2 in the attended channel in adults, whereas in children, the novelty-elicited N1 responses were faster in response to right ear stimuli and the amplitudes of the novelty-elicited N2 responses were smaller for the right ear stimuli irrespective of the direction of attention. This asymmetry in N2 amplitudes was in some degree due to an overlapping slow positivity elicited by the novel tones directed towards the right ear, which might suggest that children allocate more resources to the processing of novel stimuli presented to the right ear than to the left ear. This lateralization effect in novelty-elicited N1 and N2 in children was a new, previously unreported finding. Because no asymmetries between the ears were found with respect to the false button presses to novel stimuli, it may be considered that the aforementioned asymmetry does dot result from false processesing of novel stimuli as targets on either side. The explanation for this asymmetry may be that the two hemispheres are functionally different. The left hemisphere is verbally dominant and the right non-verbally dominant. The dichotic presentation of verbal auditory stimuli typically yields a right ear advantage (REA) for verbal and a left ear advantage (LEA) for non-verbal material when subjects are requested to report what they perceive on each trial. The standard explanation for the ear advantage is that the contralateral auditory pathways suppress the ipsilatreal pathways, thus favouring the right ear input to the left hemisphere (Kimura, 1967). An alternative view to the structural model was proposed by Kinsbourne (1970) who argued that anticipation of an incoming verbal stimulus alerted the left hemisphere for processing of the stimulus and correspondingly the non-verbal stimulus alerted the right hemisphere. Hemispheric asymmetries revealed by behavioral measures for both verbal and non-verbal material are known to correlate with ERP findings (e.g. Tenke et al., 1993; Wioland et al., 1996, 1999). Thus, the enhanced N2 in response to left ear stimuli could reflect the LEA for non-verbal stimuli, which children could not overcome. Signs of the LEA for non-verbal stimuli in children were seen also in N1 and N2 responses to standard stimuli, where the attention-elicited enhancement of the responses was seen only in the attention-left conditions. Also, the childrens' standard-elicited N1 responses were faster to the left ear stimulation in than to the right ear stimulation even if the right ear was attended. However, the absence of similar lateralizing effects target N2 or P3, together with the fact that children also can reliably overcome the ear advantage (as indicated by studies on REA) by the age of 8 or 9 years (Hiscock and Beckie, 1993), suggest that our finding may not be readily explained by the LEA only. An additional explanation for the left ear–right ear-asymmetry in children is that the right hemisphere is crucial in attention and novel cognitive situations (Goldberg et al., 1994), thus favouring the left ear input for novel unexpected tones. The exact nature of this ear-asymmetry remains to be tested. The novel P3, which is thought to reflect, orienting to new, unexpected stimuli (Knight and Scabini, 1998), did not show any attention effects in either of the groups. The novelty P3 is considered involuntary and automatic (Courchesne et al., 1975; Knight and Scabini, 1998; Squires et al., 1975), and according to the data presented here, appears not to be affected by direction of attention.

In summary, our results suggest that developmental changes occur both in utilization of early attentional selection and in target detection. In our dichotic paradigm, only children employed mechanisms of early selection. Behavioural data, on the other hand, suggests that children found the task more difficult than adults. Thus, it might be speculated that since similar task requirements may produce different perceptual load at different ages, also the employment of automatic strategies of attention may change with age. Even though children were able to employ efficiently the mechanisms of early selection when processing standard (non-novel) stimuli, their processes in relation to novel stimuli were attention-independent and even varied with ear/hemisphere. This ear-asymmetry is a previously unreported finding, and the exact nature of this finding remains to be tested. Adults, on the contrary to children, seemed to be able to maintain their attentional focus even in the presence of unexpected stimuli. Also, the processing of target stimuli seems to become more efficient with age.

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References

Bartgis J, Lilly AR, Thomas DG. Event-related potential and behavioral measures of attention in 5-, 7-, and 9-year-olds. J Gen Psychol 2003; 130(3):311–35.

Berman S, Friedman D. The development of selective attention as reflected by event-related brain potentials. J Exp Child Psychol 1995; 59(1): 1-31.

Brooker BH. The development of selective attention in learning-disabled and normal boys: an auditory evoked potential and behavioural analysis. Unpublished doctoral dissertation, Queens University, Kingston, Canada; 1980.

Bruneau N, Roux S, Guerin P, Barthelemy C, Lelord G. Temporal prominence of auditory evoked potentials (N1 wave) in 4–8-year-old children. Psychophysiology 1997;34:32–8.

Ceponiene R, Cheour M, Näätänen R. Interstimulus interval and audotory event-related potentials in children: evidence for multiple generators. Electroencephalogr Clin Neurophysiol 1998;108:345–54.

Courchesne E. Changes in P3 waves with event-repetition: long-term effects on scalp distribution and amplitude. Electroencephalogr Clin Neurophysiol 1978;45:754–66.

Courchesne E, Hillyard SA, Galambos R. Stimulus novelty, task relevance and the visual evoked potential in man. Electroencephalogr Clin Neurophysiol 1975;39:131–43.

Cycowicz YM, Friedman D. A developmental study of the effect of temporal order on the ERPs elicited by novel environmental sounds. Electroencephalogr Clin Neurophysiol 1997;103:304–18.

Cycowicz YM, Friedman D, Rothstein M. An ERP developmental study of repetition priming by auditory novel stimuli. Psychophysiology 1996; 33:680–90.

Deacon D, Breton F, Ritter W, Vaughan Jr HG. The relationship between N2 and N400: scalp distribution, stimulus probability, and task relevance. Psychophysiology 1991;28:185–200.

Dien J. Issues in the application of the average reference: review, critiques, and recommendations. Behav Res Methods, Instrum Comput 1998;3034–43.

Donald MW. Neural selectivity in auditory attention: sketch of a theory. In: Gaillard AWK, Ritter W, editors. Tutorials in event-related potential research: endogenous components. Amsterdam: Elsevier; 1983. p. 37–77.

Donchin E, Coles M. Is the P300 a manifestation of context updating? Bahav Brain Sci 1988;11:357–74.

Duncan-Johnson CC, Donchin E. On quantifying surprise: the variation in event-related potentials with subjective probability. Psychophysiology 1977;14:456–67.

Goldberg E, Podell K, Lovell M. Lateralization of frontal lobe functions and cognitive novelty. J Neuropsychiatry Clin Neurosci 1994;6:371–8.

Goodin DS, Squires KC, Henderson BH, Starr A. Age-related variations in evoked potentials to auditory stimuli in normal human subjects. Electroencephalogr Clin Neurophysiol 1978;44:447–58.

Hartley LR. The effect of stimulus relevance on the cortical evoked potentials. Q J Exp Psychol 1970;3:531–46.

Hillyard S, Hink RF, Schwent V, Picton T. Electrical signs of selective attention in human brain. Science 1973; 182(108): 177-80.

Hiscock M, Beckie JL. Overcoming the right ear advantage: a study of focused attention in children. J Clin Exp Neuropsychol 1993;5:754–72.

Johnstone SJ, Barry RJ, Anderson JW, Coyle SF. Age-related changes in child and adolescent event-related potential component morphology, amplitude and latency to standard and target stimuli in an auditory oddball task. Int J Psychophysiol 1996;24:223–38.

Kahneman D, Treisman A. Changing views of attention and automaticity. In: Parasuram R, Davies R, editors. Changing views of attention and automaticity. New York: Academic Press; 1984. p. 29–61.

Kimura D. Functional asymmetry of the brain in dichotic listening. Cortex 1967;3:163–78.

Kinsbourne M. The cerebral basis of lateral asymmetries in attention. Acta Psychol (Amst) 1970;33:193–201.

Knight RT, Scabini D. Anatomic bases of event-related potentials and their relationship to novelty detection in humans. J Clin Neurophysiol 1998; 15(1):3–13.

Korkman M, Kirk U, Kemp S. NEPSY—neuropsychological assessment of children. Finnish version. Helsinki: Psykologien kustannus; 1997.

Lane DM, Pearson DA. The development of selective attention. Merril-Palmer Q 1982;28(3):317–37.

Leppert D, Goodin DS, Aminoff MJ. Stimulus recognition and its relationship to the cerebral event-related potential. Neurology 2003; 61:1533–7.

Loiselle DL, Stamm JS, Maitinsky S, Whipple SC. Evoked potential and behavioral signs of attentive dysfunction in hyperactive boys. Psychophysiology 1980;17:193–210.

Lovrich D, Stamm JS. Event-related potential and behavioral correlates of attention in reading retardation. J Clin Neurophysiol 1983;5: 13–37.

Martin L, Barajas JJ, Fernandez R, Torres E. Auditory event-related potentials well-characterized groups of children. Electroencephalogr Clin Neurophysiol 1988;71(5):375–81.

Näätänen R. The role of attention in auditory information processing as revealed by event-related potentials and other brain measures of cognitive function. Behav Brain Sci 1990;13:201–88.

Näätänen R. Attention and brain function. Hillsdale, NJ: Lawrence Erlbaum Associates; 1992.

Näätänen R, Picton TW. N2 and automatic versus controlled processes. Electroencephalogr Clin Neurophysiol Suppl 1986;38:169–86.

Näätänen R, Picton T. The N1 wave of the human electric and magnetic response to sound: a review and analysis of the component structure. Psychophysiology 1987;24:375–425.

Näätänen R, Gaillard AW, Mäntysalo S. Early selective-attention effect on evoked potential reinterpreted. Acta Psychol (Amst) 1978;42: 313–29.

Oades RD, Dittmann-Balcar A, Zerbin D. Development and topography of auditory event-related potentials (ERPs). Mismatch and processing negativity in individuals 8–22 years of age. Psychophysiology 1997;34: 677–93.

Picton TW. The P300 wave of the human event-related potential. J Clin Neurophysiol 1992;4:456–79.

Picton T, Lins O, Scherg M. The recording and analysis of event-related poterntials. In: Boller F, Grafman (series Eds) J, Johnson (Section Ed) M, editors. Handbook of neuropsychology, vol. 10, 1980. p. 3–73 section. 14.

Polich J. Meta-analysis of P300 normative aging studies. Psychophysiology 1996;33:334–53.

Ridderinkhof KR, van der Stelt O. Attention and selection in the growing child: views derived from developmental psychophysiology. Biol Psychol 2000;54:55–106.

Ritter W, Simson R, Vaughan Jr HG, Friedman D. A brain event related to the making of a sensory discrimination. Science 1979;203: 1358–61.

Ritter W, Simson R, Vaughan Jr HG. Event-related potential correlates of two stages of information processing in physical and semantic discrimination tasks. Psychophysiology 1983;20:168–79.

Simson R, Vaughn Jr HG, Ritter W. The scalp topography of potentials in auditory and visual discrimination tasks. Electroencephalogr Clin Neurophysiol 1977;42:528–35.

Squires NK, Squires KC, Hillyard SA. Two varieties of long latency positive waves evoked by unpredictable auditory stimuli in man. Electroencephalogr Clin Neurophysiol 1975;38:387–401.

Takeshita K, Nagamine T, Thuy D, Satow T, Matsuhashi M, Yamamoto J, Takayama M, Fujiwara N, Shibasaki H. Maturational change of parallel auditory processing in school-aged children revealed by simultaneous recording of magnetic and electric cortical responses. Clin Neurophysiol 2002;113:1470.

Tenke CE, Bruder GE, Towey JP, Leite P, Sidtis JJ. Correspondence between brain ERP and behavioral asymmetries in a dichotic complex tone test. Psychophysiology 1993;30:62–70.

van der Stelt O, Kok A, Smulders FTY, Snel J, Gunning WB. Cerebral event-related potentials associated with selective attention to color: developmental changes from childhood to adulthood. Psychophysiology 1998;35:227–39.

Wechsler D. Wechsler intelligence scale for children-revised. Finnish version. Helsinki: Psykologien Kustannus; 1984.

Wioland N, Rudolf G, Metz-Lutz MN, Mutschler V, Kurtz D, Marescaux C. An electrophysiological dichotic syllable test: normative data for a right-handed population. Electroencephalogr Clin Neurophysiol Suppl 1996;46:261–70.

Wioland N, Rudolf G, Metz-Lutz MN, Mutschler V, Marescaux C. Cerebral correlates of hemispheric lateralization during a pitch discrimination task: an ERP study in dichotic situation. Clin Neurophysiol 1999;110:516–23.

Woods DL. The component structire of the N1 vawe of the human auditory evoked potential Perspect. Event-Related Potentials Res 1995;44: 102–9.